

TANVIR AHSAN SHOWROV

Mirpur, Dhaka-1207, Bangladesh

 [Google Scholar](#) |  [LinkedIn](#) |  [ResearchGate](#) |  [Portfolio](#)

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Education

Aviation and Aerospace University, Bangladesh

Jan 2022 – Mar 2026

Bachelor of Science in Avionics Engineering, **CGPA: 3.75/4.00**

Dhaka, Bangladesh

- **Class Rank: 4/30**
- **Last Semester GPA: 4.00/4.00**

Research Interests

- Biomedical Signal Processing, Deep Learning, Federated Learning, Image Processing, Applied Machine Learning, AI in Healthcare, Explainable AI, Intelligent Sensing Systems, Cyber Security

Relevant Coursework

- Signals and Systems
- Digital Signal Processing
- Data Structures and Algorithms
- Control Systems
- Machine Learning
- Radar Engineering
- Microprocessor & Microcontroller
- Real Time Embedded Systems
- Communication Systems

Research Experience

Undergraduate Thesis

Jan 2024 – Dec 2025

Supervisor: Dr. S. M. Sadakatul Bari

Department of AE, AAUB

- **Thesis Title: A Quantum-Inspired Federated Intrusion Detection System for UAV Networks with Privacy-Aware Design**
- Designed a privacy-preserving Intrusion Detection System (IDS) for UAV/FANET networks using Federated Learning (FedAvg) to enable distributed training without centralizing sensitive telemetry data.
- Developed a quantum-inspired kernel-based feature extraction pipeline with Nyström approximation, improving intrusion detection under non-IID and resource-constrained UAV environments.
- Evaluated adversarial robustness of the IDS through FGSM and PGD stress testing, analyzing resilience against evasion and transfer-style attacks.
- Integrated privacy-aware aggregation mechanisms (DP-based noise injection and secure-aggregation-compatible designs) to reduce information leakage while maintaining detection performance.
- Performed comparative experiments between centralized and federated settings, analyzing trade-offs among detection performance, privacy, communication overhead, convergence, and robustness.

Publications

- [1] Siam, M. J. N., **Showrov, T. A.**, Hossain, M. S., Ayaan, N. S., Bari, S. S., Tariq, F., Mahmud, A. A. "Comprehensive benchmarking and explainable machine learning analysis of EEG imagery activity recognition." **Scientific Reports**, 2026. (Q1 journal; Nature Portfolio). DOI: [10.1038/s41598-026-50997-y](https://doi.org/10.1038/s41598-026-50997-y)
- [2] **T. A. Showrov**, M. J. N. Siam, N. S. Ayaan, M. S. Hossain and S. M. S. Bari, "Robust Epileptic Seizure Detection via Multi-Artifact EEG Denoising and SE-CNN," 28th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2025, pp. 1666–1671, DOI: [10.1109/ICCIT68739.2025.11489540](https://doi.org/10.1109/ICCIT68739.2025.11489540)

Projects

FM Transmitter Design and Simulation

Oct 2024 – Nov 2024

| *Communication Systems Sessional* | Supervised By: Dr. Md. Sakir Hossain

- Simulated an FM transmitter using MATLAB and Simulink to study frequency modulation principles, including carrier wave modulation and noise resistance.
- Implemented a practical FM transmitter using EasyEDA with components like a microphone, LM386N amplifier, 24 MHz crystal oscillator, and antenna to achieve wireless communication.

Orthogonal Frequency Division Multiplexing (OFDM)

Sept 2024 – Nov 2024

| *Communication Systems* | Supervised By: Dr. Md. Sakir Hossain

- Simulated and analyzed an OFDM communication system in MATLAB, enabling efficient data transmission by dividing data into orthogonal subcarriers.

- Simulated key parameters such as Bit Error Rate (BER) and Signal-to-Noise Ratio (SNR) in MATLAB, validating system performance under noise and interference.
- Investigated resilience to multipath fading and implemented FFT/IFFT for efficient signal processing.

Smart Traffic Control Signal System

Apr 2023 – Jun 2023

| *Digital Logic & Computer Design* | Supervised By: Dr. Md. Sakir Hossain

- Developed a density-based smart traffic control system using Arduino UNO and IR sensors to dynamically adjust signal timings, reducing traffic congestion and delays.
- Simulated the system in Proteus and implemented a functional prototype with IC74HC595 shift registers, IC74HC08 AND gates, and LEDs.

Extracurricular Activities

- Vice President, AAUB Debating Club (2024 & 2025)
- Vice President, AAUB Career Club (2025)
- Director (Publication, Communication & Outreach), AAUB AI & Robotics Club (2023 & 2024)
- Finance Secretary, AAUB Adventure & Photography Club (2024 & 2025)
- Web Manager, AAUB Sports Club (2024 & 2025)

Technical Skills

Programming Languages: C, C++, Python

Python/ML Libraries: NumPy, Pandas, Scikit-learn, TensorFlow/PyTorch, Matplotlib

Simulation Software: MATLAB Simulink, SolidWorks Simulation

Data/Visualization Tool: OriginLab, Tecplot 360

Documentation & Productivity: Microsoft Office Suite, LaTeX

Design Software: Adobe Photoshop, Adobe Illustrator, Canva

Achievements & Awards

- Pro-Vice Chancellor's Scholarship for academic excellence (Jan 2024, Jun 2024, Jan 2025 & Jun 2025 Semesters)
- University Merit Scholarship (Jun 2021 & Jan 2022 Semesters)
- 3rd Place, Secondary Category, Inter School Annual Science Fair, Sher-e-Bangla Nagar Govt. Boys' High School (2016)

Industrial Experience

Radar & CNS Facilities, Hazrat Shahjalal International Airport (HSIA)

Nov 2025

Industrial Visit

Dhaka, Bangladesh

- Observed the operational architecture of primary and secondary surveillance radar (PSR/SSR) systems used for civil air traffic management at Bangladesh's busiest international airport.
- Gained exposure to Communication, Navigation, and Surveillance (CNS) infrastructure, including radar data processing, airspace monitoring, and aircraft tracking for safe and efficient air traffic control operations.

208 MOU & 216 MRO Unit, Bangladesh Air Force (BAF) Base Bashar

Jun 2024 – Jul 2024

Industrial Trainee

Dhaka, Bangladesh

- Hands-on experience with the maintenance and overhauling of Bell 206, Bell 212, and MI-17 helicopters in 208 MOU and 216 MRO units.
- Collaborated with teams to observe and participate in repair, maintenance, and overhauling operations, ensuring operational readiness of the aircraft.

Certifications

- Coursera – Machine Learning Specialization by DeepLearning.AI & Stanford University; Excel Skills for Business: Essentials

References

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